

CometCAD: From Symbol To PCB Layout

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CometCAD is a tool for creating circuit schematics and printed circuit boards (PCBs). With the tiny green board inside every electronic system deciding the complete functionality of the system, creating worthy designs take priority. With this thought in mind, this month we explore designing a PCB using CometCAD.

CometCAD is built for Windows and falls under free-use licence. Design of the PCB using this software is divided into three stages: symbols, circuit and layout. Let us take a look at how each of these translate to the expected end design we are looking for. To begin with, we have a few notable features.

Look at your drawings how you want

As you work with CometCAD, you have the option to customise the settings of the screen. You might be working on a tiny-screen laptop or a wide-screen

desktop, but when dealing with design, the correct zoom and optimum aspect ratio turn out to be as important as the design itself. You also have the liberty to configure the workspace for your circuit, right from the size of the sheet to spacing and visibility of grid lines.

Leave behind a footprint

A footprint is a pattern describing the arrangement of pads or through-holes used to physically attach and electrically connect the components to a PCB. Effectively, it is similar to the map of the components on the board. With CometCAD, symbols

can be linked to footprints, while off-page connectors can contain footprints other than for those that work as signal connectors.

The connectors then refer to the footprint for their pin configuration, which can be accessed individually, or the total number of pins can be split between sheets. Once created, symbols and footprints are stored in the library for further use. These are termed parts and can simply be added to the circuit like any ordinary symbol.

Fit in your PCB anywhere, anyhow

Often we see PCBs fitted onto other boards with nails or screws. Holes for these can be added in the design stage itself, via the tool. But the allowed size of the hole is fixed. If holes of other dimensions are needed, at design stage you can add a polygon around the region, so as not to route traces through that area.

Specifications you should know

- Number of circuit schematic sheets: 2
- Maximum number of symbols per sheet: 50
- Number of pins per symbol: 999
- Number of symbols in the library file: 2000
- Size of the PCB: 102mm x 102mm
- PCB panel size: 210mm x 210mm
- Number of copper layers: 2
- Maximum number of package pins: 250

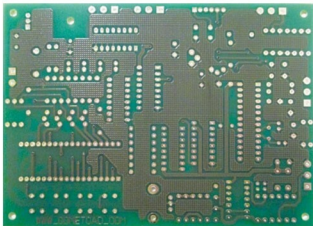
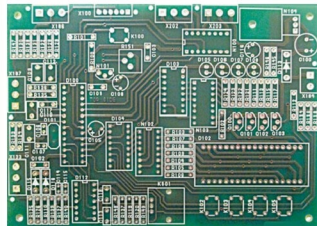


Fig. 1: Sample of a dual-layer PCB